

Stein's Method, Learning, and Inference

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Motivation: Large-scale Posterior Inference

Example: Bayesian logistic regression

- 1 Feature vectors: $v_l \in \mathbb{R}^d$ for each datapoint $l = 1, \dots, L$
 - 2 Binary class labels: $Y_l \in \{0, 1\}$, $\mathbb{P}(Y_l = 1 \mid v_l, \beta) = \frac{1}{1 + e^{-\langle \beta, v_l \rangle}}$
 - 3 Unknown parameter vector: $\beta \sim \mathcal{N}(0, I)$
- Generative model simple to express
 - Posterior distribution over unknown parameters is **complex**
 - Normalization constant **unknown**, exact integration **intractable**

v_1	v_2	v_3	v_4
3	1	0	9
0	6	8	4
5	0	3	1
2	9	7	0
Class 0		Class 1	

Standard inferential approach: Use Markov chain Monte Carlo (MCMC) to (eventually) draw samples from the posterior distribution

- **Benefit:** Approximates intractable posterior expectations $\mathbb{E}_P[h(Z)] = \int p(x)h(x)dx$ with asymptotically exact sample estimates $\mathbb{E}_Q[h(X)] = \frac{1}{n} \sum_{i=1}^n h(x_i)$
- **Problem:** Each new MCMC sample point x_i requires iterating over entire observed dataset: **prohibitive** when dataset is large!

Motivation: Large-scale Posterior Inference

Question: How do we scale Markov chain Monte Carlo (MCMC) posterior inference to massive datasets?

- **Benefit:** Approximates intractable posterior expectations $\mathbb{E}_P[h(Z)] = \int p(x)h(x)dx$ with asymptotically exact sample estimates $\mathbb{E}_Q[h(X)] = \frac{1}{n} \sum_{i=1}^n h(x_i)$
- **Problem:** Each point x_i requires iterating over entire dataset!

Template solution: Approximate MCMC with subset posteriors

[Welling and Teh, 2011, Ahn, Korattikara, and Welling, 2012, Korattikara, Chen, and Welling, 2014]

- Approximate standard MCMC procedure in a manner that makes use of only a small subset of datapoints per sample
- Reduced computational overhead leads to faster sampling and **reduced Monte Carlo variance**
- Introduces **asymptotic bias**: target distribution is not stationary
- Hope that for fixed amount of sampling time, variance reduction will outweigh bias introduced

Motivation: Large-scale Posterior Inference

Template solution: Approximate MCMC with subset posteriors

[Welling and Teh, 2011, Ahn, Korattikara, and Welling, 2012, Korattikara, Chen, and Welling, 2014]

- Hope that for fixed amount of sampling time, variance reduction will outweigh bias introduced

Introduces new challenges

- How do we compare and evaluate samples from approximate MCMC procedures?
- How do we select samplers and their tuning parameters?
- How do we quantify the bias-variance trade-off explicitly?

Difficulty: Standard evaluation criteria like effective sample size, trace plots, and variance diagnostics **assume convergence to the target distribution** and **do not account for asymptotic bias**

This talk: Introduce new quality measures suitable for comparing the quality of approximate MCMC samples

Quality Measures for Samples

Challenge: Develop measure suitable for comparing the quality of *any* two samples approximating a common target distribution

Given

- **Continuous target distribution** P with support $\mathcal{X} = \mathbb{R}^d$ and density p
 - p known up to normalization, **integration under P is intractable**
- **Sample points** $x_1, \dots, x_n \in \mathcal{X}$
 - Define **discrete distribution** Q_n with, for any function h ,
 $\mathbb{E}_{Q_n}[h(X)] = \frac{1}{n} \sum_{i=1}^n h(x_i)$ used to approximate $\mathbb{E}_P[h(Z)]$
 - We make no assumption about the provenance of the x_i

Goal: Quantify how well $\mathbb{E}_{Q_n}$ approximates $\mathbb{E}_P$ in a manner that

- I. Detects when a sample sequence **is converging** to the target
- II. Detects when a sample sequence **is not converging** to the target
- III. Is **computationally feasible**

Integral Probability Metrics

Goal: Quantify how well $\mathbb{E}_{Q_n}$ approximates $\mathbb{E}_P$

Idea: Consider an **integral probability metric (IPM)** [Müller, 1997]

$$d_{\mathcal{H}}(Q_n, P) = \sup_{h \in \mathcal{H}} |\mathbb{E}_{Q_n}[h(X)] - \mathbb{E}_P[h(Z)]|$$

- Measures maximum discrepancy between sample and target expectations over a class of real-valued test functions $\mathcal{H}$
- When $\mathcal{H}$ sufficiently large, convergence of $d_{\mathcal{H}}(Q_n, P)$ to zero implies $(Q_n)_{n \geq 1}$ converges weakly to P (Requirement II)

Problem: Integration under P intractable!

$\Rightarrow$ Most IPMs cannot be computed in practice

Idea: Only consider functions with $\mathbb{E}_P[h(Z)]$ known *a priori* to be 0

- Then IPM computation only depends on Q_n !
- How do we select this class of test functions?
- Will the resulting discrepancy measure track sample sequence convergence?
- How do we solve the resulting optimization problem in practice?

Stein's Method

Stein's method [1972] provides a recipe for controlling convergence:

- 1 **Identify operator $\mathcal{T}$ and set $\mathcal{G}$** of functions $g : \mathcal{X} \rightarrow \mathbb{R}^d$ with

$$\mathbb{E}_P[(\mathcal{T}g)(Z)] = 0 \quad \text{for all } g \in \mathcal{G}.$$

$\mathcal{T}$ and $\mathcal{G}$ together define the **Stein discrepancy** [Gorham and Mackey, 2015]

$$\mathcal{S}(Q_n, \mathcal{T}, \mathcal{G}) \triangleq \sup_{g \in \mathcal{G}} |\mathbb{E}_{Q_n}[(\mathcal{T}g)(X)]| = d_{\mathcal{T}\mathcal{G}}(Q_n, P),$$

an IPM-type measure with no explicit integration under P

- 2 **Lower bound $\mathcal{S}(Q_n, \mathcal{T}, \mathcal{G})$ by reference IPM $d_{\mathcal{H}}(Q_n, P) \Rightarrow (Q_n)_{n \geq 1}$**
converges to P whenever $\mathcal{S}(Q_n, \mathcal{T}, \mathcal{G}) \rightarrow 0$ ([Requirement II](#))

- Performed once, in advance, for large classes of distributions

- 3 **Upper bound $\mathcal{S}(Q_n, \mathcal{T}, \mathcal{G})$ by any means necessary** to demonstrate convergence to 0 ([Requirement I](#))

Standard use: As analytical tool to prove convergence

Our goal: Develop Stein discrepancy into practical quality measure

Identifying a Stein Operator $\mathcal{T}$

Goal: Identify operator $\mathcal{T}$ for which $\mathbb{E}_P[(\mathcal{T}g)(Z)] = 0$ for all $g \in \mathcal{G}$

Approach: Generator method of Barbour [1988, 1990], Götze [1991]

- Identify a Markov process $(Z_t)_{t \geq 0}$ with stationary distribution P
- Under mild conditions, its **infinitesimal generator**

$$(\mathcal{A}u)(x) = \lim_{t \rightarrow 0} (\mathbb{E}[u(Z_t) \mid Z_0 = x] - u(x))/t$$

satisfies $\mathbb{E}_P[(\mathcal{A}u)(Z)] = 0$

Overdamped Langevin diffusion: $dZ_t = \frac{1}{2} \nabla \log p(Z_t) dt + dW_t$

- Generator: $(\mathcal{A}_P u)(x) = \frac{1}{2} \langle \nabla u(x), \nabla \log p(x) \rangle + \frac{1}{2} \langle \nabla, \nabla u(x) \rangle$
- **Stein operator:** $(\mathcal{T}_P g)(x) \triangleq \langle g(x), \nabla \log p(x) \rangle + \langle \nabla, g(x) \rangle$

[Gorham and Mackey, 2015, Oates, Girolami, and Chopin, 2016]

- Depends on P only through $\nabla \log p$; computable even if p cannot be normalized!
- $\mathbb{E}_P[(\mathcal{T}_P g)(Z)] = 0$ for all $g : \mathcal{X} \rightarrow \mathbb{R}^d$ in **classical Stein set**

$$\mathcal{G}_{\|\cdot\|} = \left\{ g : \sup_{x \neq y} \max \left(\|g(x)\|^*, \|\nabla g(x)\|^*, \frac{\|\nabla g(x) - \nabla g(y)\|^*}{\|x - y\|} \right) \leq 1 \right\}$$

Detecting Convergence and Non-convergence

Goal: Show **classical Stein discrepancy** $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|}) \rightarrow 0$ if and only if $(Q_n)_{n \geq 1}$ converges to P

- In the univariate case ($d = 1$), known that for many targets P ,
 $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|}) \rightarrow 0$ only if Wasserstein $d_{\mathcal{W}_{\|\cdot\|}}(Q_n, P) \rightarrow 0$

[Stein, Diaconis, Holmes, and Reinert, 2004, Chatterjee and Shao, 2011, Chen, Goldstein, and Shao, 2011]

- Few multivariate targets have been analyzed (see [Reinert and Röllin, 2009, Chatterjee and Meckes, 2008, Meckes, 2009] for multivariate Gaussian)

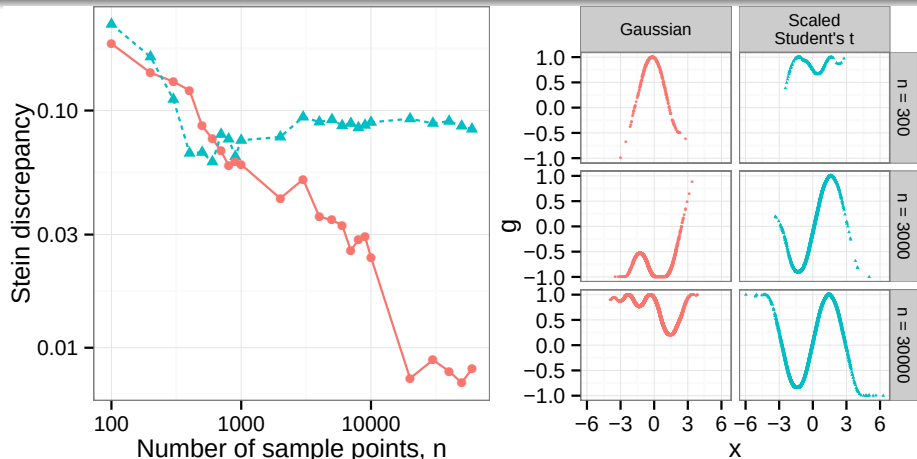
New contribution [Gorham, Duncan, Vollmer, and Mackey, 2019]

Theorem (Stein Discrepancy-Wasserstein Equivalence)

If the Langevin diffusion couples at an integrable rate and $\nabla \log p$ is Lipschitz, then
 $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|}) \rightarrow 0 \Leftrightarrow d_{\mathcal{W}_{\|\cdot\|}}(Q_n, P) \rightarrow 0.$

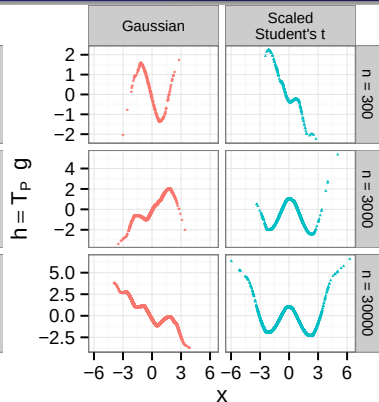
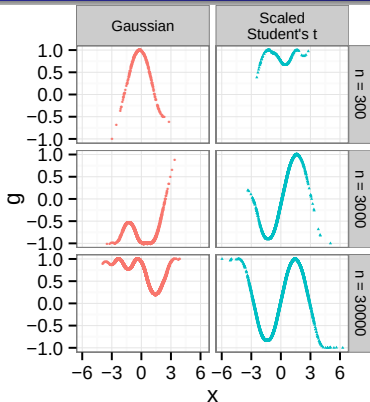
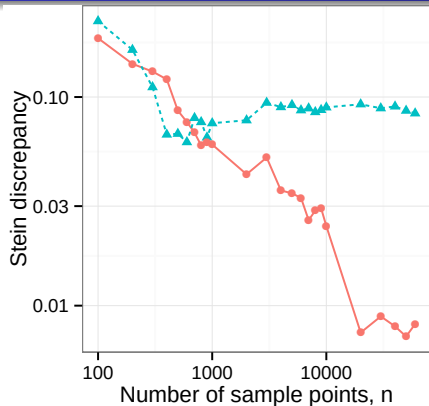
- Examples: strongly log concave P , Bayesian logistic regression or robust t regression with Gaussian priors, Gaussian mixtures
- Conditions not necessary: template for bounding $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|})$

A Simple Example



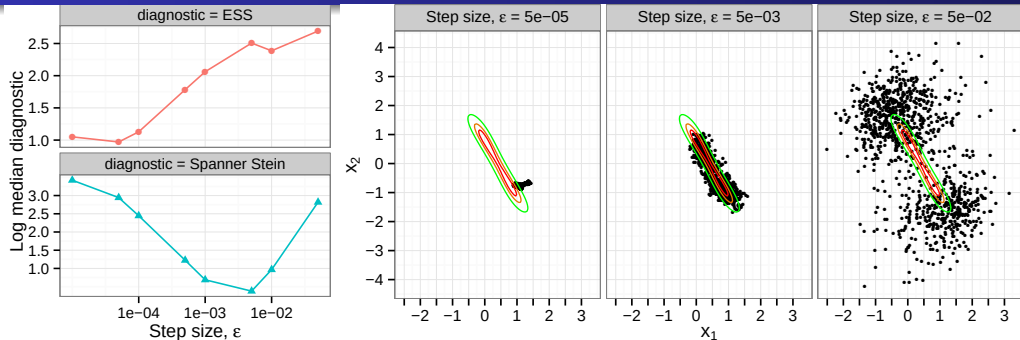
- For target $P = \mathcal{N}(0, 1)$, compare i.i.d. $\mathcal{N}(0, 1)$ sample sequence $Q_{1:n}$ to scaled Student's t sequence $Q'_{1:n}$ with matching variance
- Expect $\mathcal{S}(Q_{1:n}, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|, Q, G_1}) \rightarrow 0$ & $\mathcal{S}(Q'_{1:n}, \mathcal{T}_P, \mathcal{G}_{\|\cdot\|, Q, G_1}) \not\rightarrow 0$

A Simple Example



- **Middle:** Recovered optimal functions g
- **Right:** Associated test functions $h(x) \triangleq (\mathcal{T}_P g)(x)$ which best discriminate sample Q from target P

Selecting Sampler Hyperparameters



Target posterior density: $p(x) \propto \pi(x) \prod_{l=1}^L \pi(y_l | x)$

Stochastic Gradient Langevin Dynamics [Welling and Teh, 2011]

$$x_{k+1} \sim \mathcal{N}(x_k + \frac{\epsilon}{2}(\nabla \log \pi(x_k) + \frac{L}{|\mathcal{B}_k|} \sum_{l \in \mathcal{B}_k} \nabla \log \pi(y_l | x_k)), \epsilon I)$$

- Random batch $\mathcal{B}_k$ of datapoints used to draw each sample point
 - Step size ϵ too small $\Rightarrow$ **slow mixing**
 - Step size ϵ too large $\Rightarrow$ **sampling from very different distribution**
 - Standard diagnostics like **effective sample size (ESS)** do not account for this bias

Alternative Stein Sets $\mathcal{G}$

Goal: Identify a more “user-friendly” Stein set $\mathcal{G}$ than the classical

Approach: Reproducing kernels $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ [Oates, Girolami, and Chopin, 2016,

Chwialkowski, Strathmann, and Gretton, 2016, Liu, Lee, and Jordan, 2016]

- A reproducing kernel k is **symmetric** ($k(x, y) = k(y, x)$) and **positive semidefinite** ($\sum_{i,l} c_i c_l k(z_i, z_l) \geq 0, \forall z_i \in \mathcal{X}, c_i \in \mathbb{R}$)

- Gaussian: $k(x, y) = e^{-\frac{1}{2}\|x-y\|_2^2}$, IMQ: $k(x, y) = \frac{1}{(1+\|x-y\|_2^2)^{1/2}}$

- Generates a reproducing kernel Hilbert space (RKHS) $\mathcal{K}_k$

- Define the **kernel Stein set** [Gorham and Mackey, 2017]

$$\mathcal{G}_k \triangleq \{g = (g_1, \dots, g_d) \mid \|v\|^* \leq 1 \text{ for } v_j \triangleq \|g_j\|_{\mathcal{K}_k}\}$$

- Yields **closed-form kernel Stein discrepancy (KSD)**

$$\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) = \|w\| \text{ for } w_j \triangleq \sqrt{\sum_{i,i'=1}^n k_0^j(x_i, x_{i'})}.$$

- Reduces to **parallelizable** pairwise evaluations of **Stein kernels**

$$k_0^j(x, y) \triangleq \frac{1}{p(x)p(y)} \nabla_{x_j} \nabla_{y_j} (p(x)k(x, y)p(y))$$

Detecting Non-convergence

Goal: Show $(Q_n)_{n \geq 1}$ converges to P whenever $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$

Theorem (Univariate KSD detects non-convergence [Gorham and Mackey, 2017])

Suppose $P \in \mathcal{P}$ and $k(x, y) = \Phi(x - y)$ for $\Phi \in C^2$ with a non-vanishing generalized Fourier transform. If $d = 1$, then $(Q_n)_{n \geq 1}$ converges weakly to P whenever $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$.

- $\mathcal{P}$ is the set of targets P with Lipschitz $\nabla \log p$ and strongly log concave tails
($\frac{\langle \nabla \log(p(x)/p(y)), y-x \rangle}{\|x-y\|_2^2} \geq k$ for $\|x-y\|_2 \geq r$)
 - Includes Bayesian logistic and Student's t regression with Gaussian priors, Gaussian mixtures with common covariance, ...
- Justifies use of KSD with popular Gaussian, Matérn, or inverse multiquadric kernels **k in the univariate case**

Detecting Non-convergence

Goal: Show $(Q_n)_{n \geq 1}$ converges to P whenever $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$

- In higher dimensions, KSDs based on common kernels **fail to detect non-convergence**, even for Gaussian targets P

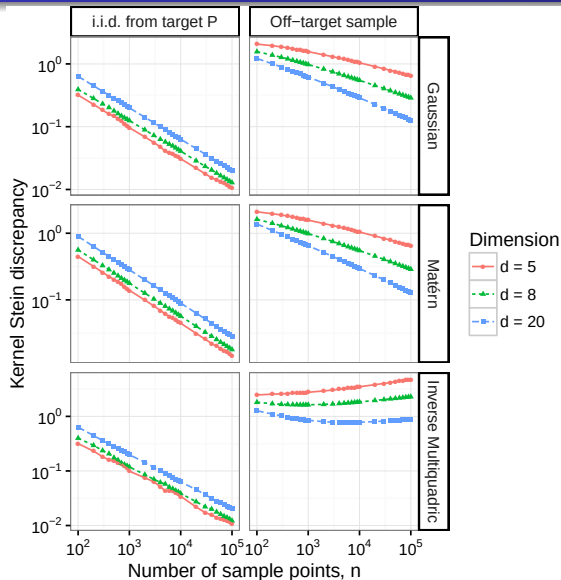
Theorem (KSD fails with light kernel tails [Gorham and Mackey, 2017])

*Suppose $d \geq 3$, $P = \mathcal{N}(0, I_d)$, and $\alpha \triangleq (\frac{1}{2} - \frac{1}{d})^{-1}$. If $k(x, y)$ and its derivatives decay at a $o(\|x - y\|_2^{-\alpha})$ rate as $\|x - y\|_2 \rightarrow \infty$, then $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$ for some $(Q_n)_{n \geq 1}$ **not converging** to P .*

- Gaussian ($k(x, y) = e^{-\frac{1}{2}\|x-y\|_2^2}$) and Matérn kernels fail for $d \geq 3$
- Inverse multiquadric kernels ($k(x, y) = (1 + \|x - y\|_2^2)^\beta$) with $\beta < -1$ fail for $d > \frac{2\beta}{1+\beta}$
- The violating sample sequences $(Q_n)_{n \geq 1}$ are simple to construct

Problem: Kernels with light tails ignore excess mass in the tails

The Importance of Kernel Choice



- KSDs with light-tailed kernels **fail to detect non-convergence** when $d \geq 3$!
- Target $P = \mathcal{N}(0, I_d)$
- Off-target Q_n has all $\|x_i\|_2 \leq 2n^{1/d} \log n$, $\|x_i - x_j\|_2 \geq 2 \log n$
- Gaussian and Matérn KSDs driven to 0 by an off-target sequence that does not converge to P
- IMQ KSD does not have this deficiency: $k(x, y) = (1 + \|x - y\|_2^2)^{-\frac{1}{2}}$

Detecting Non-convergence

Goal: Show $(Q_n)_{n \geq 1}$ converges to P whenever $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$

- Consider the inverse multiquadric (IMQ) kernel

$$k(x, y) = (c^2 + \|x - y\|_2^2)^\beta \text{ for some } \beta < 0, c \in \mathbb{R}.$$

- IMQ KSD **fails to detect non-convergence** when $\beta < -1$
- However, IMQ KSD **detects non-convergence** when $\beta \in (-1, 0)$

Theorem (IMQ KSD detects non-convergence [Gorham and Mackey, 2017])

Suppose $P \in \mathcal{P}$ and $k(x, y) = (c^2 + \|x - y\|_2^2)^\beta$ for $\beta \in (-1, 0)$. If $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$, then $(Q_n)_{n \geq 1}$ converges weakly to P .

Detecting Convergence

Goal: Show $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$ whenever $(Q_n)_{n \geq 1}$ converges to P

Proposition (KSD detects convergence [Gorham and Mackey, 2017])

If $k \in C_b^{(2,2)}$ and $\nabla \log p$ Lipschitz and square integrable under P , then $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$ whenever the Wasserstein distance $d_{\mathcal{W}_{\|\cdot\|_2}}(Q_n, P) \rightarrow 0$.

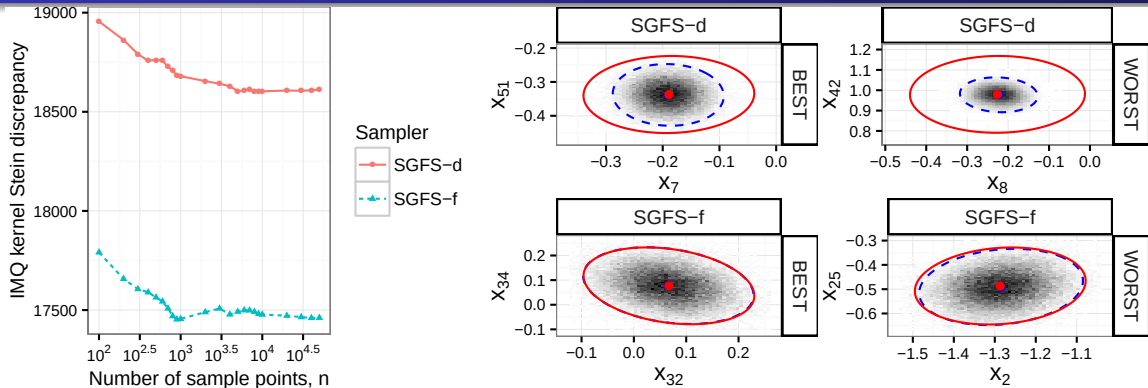
- Covers Gaussian, Matérn, IMQ, and other common bounded kernels k

Stochastic Gradient Fisher Scoring (SGFS)

[Ahn, Korattikara, and Welling, 2012]

- Approximate MCMC procedure designed for scalability
 - Approximates Metropolis-adjusted Langevin algorithm but does not use Metropolis-Hastings correction
 - Target P is not stationary distribution
- **Goal:** Choose between two variants
 - SGFS-f inverts a $d \times d$ matrix for each new sample point
 - SGFS-d inverts a diagonal matrix to reduce sampling time
- **MNIST handwritten digits** [Ahn, Korattikara, and Welling, 2012]
 - 10000 images, 51 features, binary label indicating whether image of a 7 or a 9
- Bayesian logistic regression posterior P

Selecting Samplers



- **Left:** IMQ KSD quality comparison for SGFS Bayesian logistic regression (no surrogate ground truth used)
- **Right:** SGFS sample points (5×10^4) with marginal means and 95% confidence ellipses (blue) that align best / worst with surrogate ground truth sample (red)
- Small speed-up of SGFS-d (0.0017s vs. 0.0019s) outweighed by loss in accuracy

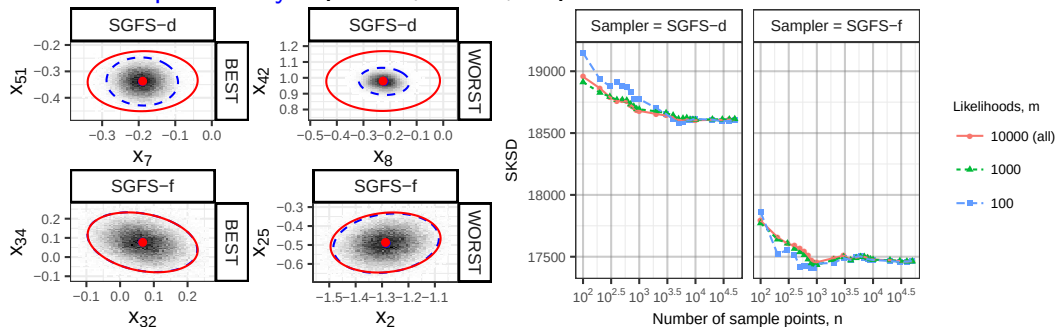
Stochastic Stein Discrepancies

Issue: What if $\nabla \log p$ is too expensive to evaluate?

- Posterior $\nabla \log p(x) = \nabla \log \pi(x) + \sum_{l=1}^L \nabla \log \pi(y_l | x)$

Solution: Stochastic Stein Discrepancies [Gorham, Raj, and Mackey, 2020]

- Replace each $\nabla \log p(x_i)$ with stochastic gradient based on random datapoint batch: $\nabla \log \pi(x_i) + \frac{L}{|\mathcal{B}_i|} \sum_{l \in \mathcal{B}_i} \nabla \log \pi(y_l | x_i)$
- Resulting stochastic Stein discrepancies **inherit convergence control** of standard SDs **with probability 1** [Gorham, Raj, and Mackey, 2020]



Beyond Sample Quality Comparison

Goodness-of-fit testing

- Chwialkowski, Strathmann, and Gretton [2016] used the KSD $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k)$ to test whether a sample was drawn from a target distribution P (see also Liu, Lee, and Jordan [2016])
- Test with default Gaussian kernel k experienced considerable loss of power as the dimension d increased
- We recreate their experiment with IMQ kernel ($\beta = -\frac{1}{2}, c = 1$)
 - For $n = 500$, generate sample $(x_i)_{i=1}^n$ with $x_i = z_i + u_i e_1$ $z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$ and $u_i \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]$. Target $P = \mathcal{N}(0, I_d)$.
 - Compare with standard normality test of Baringhaus and Henze [1988]

Table: Mean power of multivariate normality tests across 400 simulations

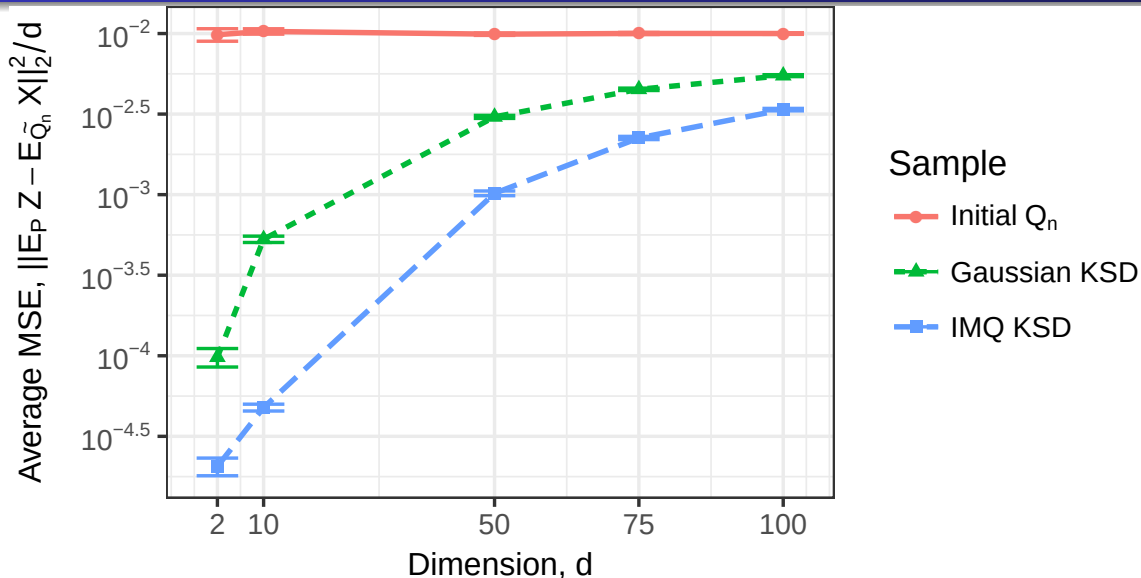
	d=2	d=5	d=10	d=15	d=20	d=25
B&H	1.0	1.0	1.0	0.91	0.57	0.26
Gaussian	1.0	1.0	0.88	0.29	0.12	0.02
IMQ	1.0	1.0	1.0	1.0	1.0	1.0

Beyond Sample Quality Comparison

Improving sample quality

- Given sample points $(x_i)_{i=1}^n$, can minimize KSD $\mathcal{S}(\tilde{Q}_n, \mathcal{T}_P, \mathcal{G}_k)$ over all weighted samples $\tilde{Q}_n = \sum_{i=1}^n q_n(x_i) \delta_{x_i}$ for q_n a probability mass function
- Liu and Lee [2017] do this with Gaussian kernel $k(x, y) = e^{-\frac{1}{h} \|x-y\|_2^2}$
 - Bandwidth h set to median of the squared Euclidean distance between pairs of sample points
- We recreate their experiment with the IMQ kernel $k(x, y) = (1 + \frac{1}{h} \|x - y\|_2^2)^{-1/2}$

Improving Sample Quality



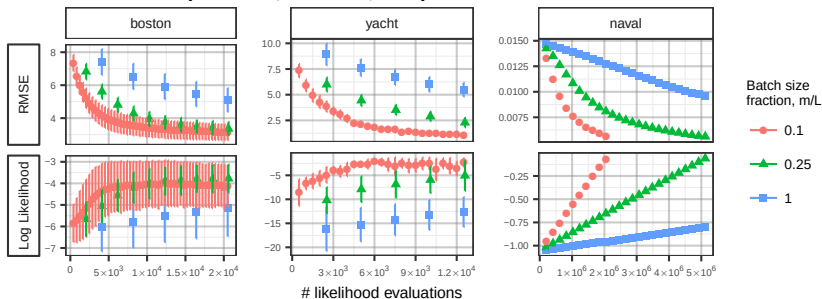
Generating High-quality Samples

Stein Variational Gradient Descent (SVGD) [Liu and Wang, 2016]

- Uses KSD to repeatedly update locations of n sample points:

$$x_i \leftarrow x_i + \frac{\epsilon}{n} \sum_{l=1}^n (k(x_l, x_i) \nabla \log p(x_l) + \nabla_{x_l} k(x_l, x_i))$$

- Approximates gradient step in KL divergence
- Drives KSD to 0 at $O(1/\sqrt{n})$ rate [Balasubramanian, Banerjee, and Ghosal, 2024]
- Simple to implement (but each update costs n^2 time)
- **Stochastic SVGD:** uses stochastic KSD $\Rightarrow$ same guarantees with many fewer likelihood evaluations [Gorham, Raj, and Mackey, 2020]



Generating High-quality Samples

Stein Points [Chen, Mackey, Gorham, Briol, and Oates, 2018]

- Greedily minimizes KSD by constructing $Q_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ with

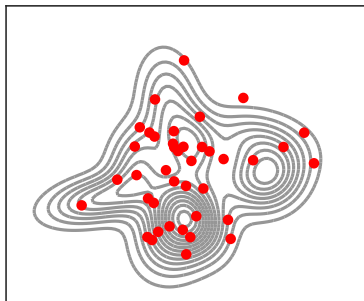
$$x_n \in \operatorname{argmin}_x \mathcal{S}\left(\frac{n-1}{n}Q_{n-1} + \frac{1}{n}\delta_x, \mathcal{T}_P, \mathcal{G}_k\right) = \operatorname{argmin}_x \sum_{j=1}^d \frac{k_0^j(x, x)}{2} + \sum_{i=1}^{n-1} k_0^j(x_i, x)$$

- Sends KSD to zero at $O(\sqrt{\log(n)/n})$ rate

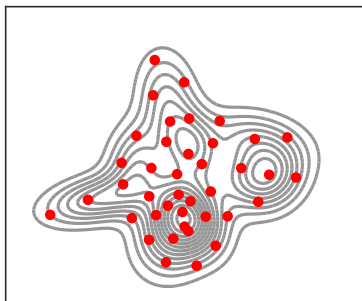
Stein Point MCMC [Chen, Barp, Briol, Gorham, Girolami, Mackey, and Oates, 2019]

- Suffices to optimize over iterates of a Markov chain

MCMC



SP-MCMC

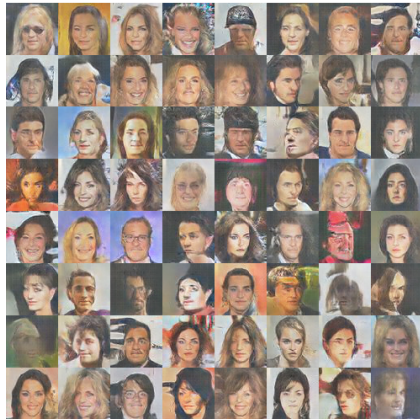


Many opportunities for future development

- ① Improving scalability while maintaining convergence control
 - Subsampling of likelihood terms in $\nabla \log p$ [Gorham, Raj, and Mackey, 2020]
 - Linear time, low-rank kernels that distinguish distributions
 - **Finite set Stein discrepancies** [Jitkrittum, Xu, Szabó, Fukumizu, and Gretton, 2017]
 - **Random feature Stein discrepancies** [Huggins and Mackey, 2018]
 - **Open question:** When do such discrepancies control convergence?
- ② Exploring the impact of Stein operator choice
 - An infinite number of operators $\mathcal{T}$ characterize P
 - **Open questions:** How is discrepancy impacted? How do we select the best $\mathcal{T}$?
 - **Heavy tails:** If $\nabla \log p$ bounded and $k \in C_0^{(1,1)}$, KSD **does not** control convergence
 - **Diffusion Stein operators** $(\mathcal{T}g)(x) = \frac{1}{p(x)} \langle \nabla, p(x)a(x)g(x) \rangle$ of Gorham, Duncan, Vollmer, and Mackey [2019] may be appropriate for such heavy-tailed distributions
 - **Isolated modes:** Langevin SDs struggle to detect **unexplored modes**. Better $\mathcal{T}$?

Training generative models

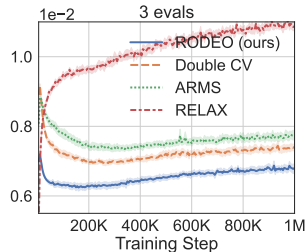
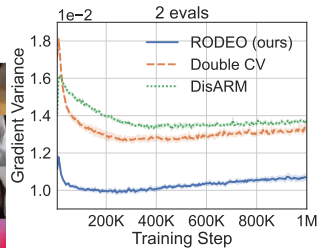
[Wang and Liu, 2016, Pu, Gan, Henao, Li, Han, and Carin, 2017, Shi, Zhou, Hwang, Titsias, and Mackey, 2022b]



DCGAN



SteinGAN



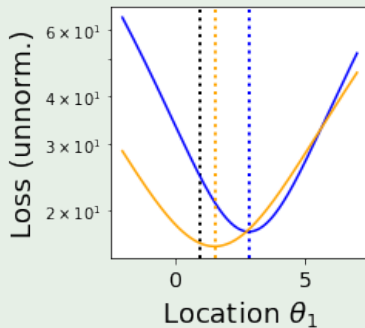
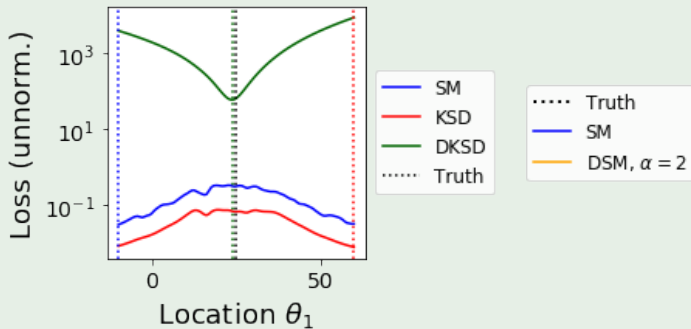
Future Directions

Parameter estimation in unnormalized models

Example (Minimum Stein Discrepancy Estimation [Barp, Briol, Duncan, Girolami, and Mackey, 2019])

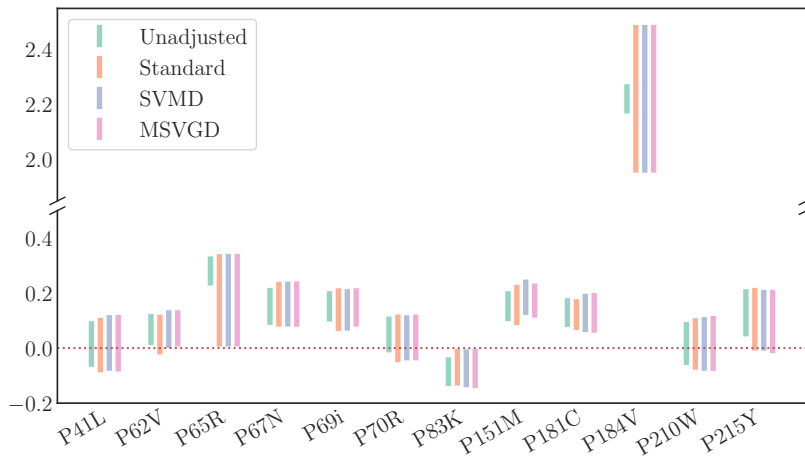
$$\hat{\theta} \in \operatorname{argmin}_{\theta \in \Theta} \mathcal{S}(Q_n, \mathcal{T}_{P_\theta}, \mathcal{G})$$

- Unlike maximum likelihood, **avoids normalization constant / integration under P_θ !**
- Can design diffusion-based discrepancies to deal with heavy tails and outliers



Future Directions

Post-selection inference



- **Constrained targets**
 P arise when testing significance after variable selection

[Tian and Taylor, 2018]

- **Stein Variational Mirror Descent** and **Mirrored SVGD** can derive confidence intervals for constrained P

[Shi, Liu, and Mackey, 2022a]

Non-convex optimization

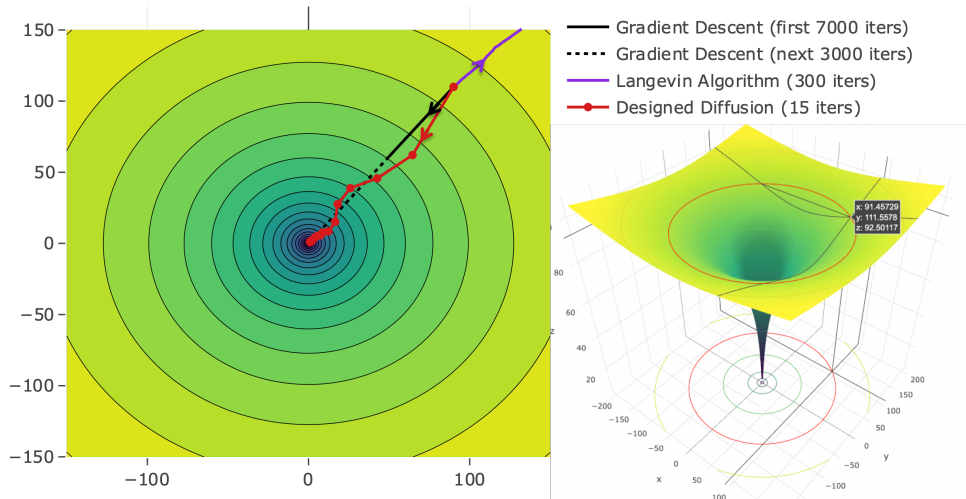
Example (Optimization with Discretized Diffusions [Erdogdu, Mackey, and Shamir, 2018])

- To minimize $f(x)$, choose $a(x) \succcurlyeq cI$ with $a(x)\nabla f(x)$ Lipschitz and **distantly dissipative** ($\frac{\langle a(x)\nabla f(x) - a(y)\nabla f(y), x - y \rangle}{\|x - y\|_2^2} \geq k$ for $\|x - y\|_2 \geq r$)
- Approximate target sequence $p_n(x) \propto e^{-\gamma_n f(x)}$ using Markov chain $x_{n+1} \sim \mathcal{N}(x_n - \frac{\epsilon_n}{2} a(x_n) \nabla f(x_n) + \frac{\epsilon_n}{2\gamma_n} \langle \nabla, a(x_n) \rangle, \frac{\epsilon_n}{\gamma_n} a(x_n))$
- **Thm:** $\min_{1 \leq i \leq n} \mathbb{E} f(x_i) \rightarrow \min_x f(x)$ (with explicit error bounds) for appropriate ϵ_n and γ_n when ∇f , ∇a , and $a^{1/2}$ are Lipschitz

Future Directions

Non-convex optimization [Erdogdu, Mackey, and Shamir, 2018]

$$\min_x f(x) = 5 \log(1 + \frac{1}{2} \|x\|_2^2), \quad a(x) = (1 + \frac{1}{2} \|x\|_2^2)I, \quad a(x)\nabla f(x) = 5x$$

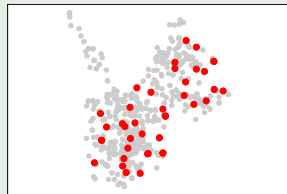
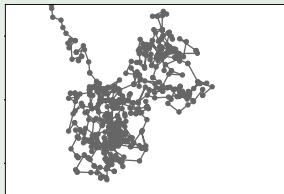
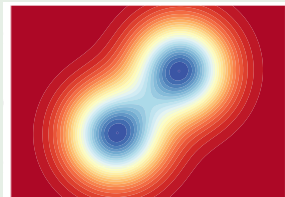


Future Directions

Distribution compression

Example (Stein Kernel Thinning [Li, Dwivedi, and Mackey, 2024])

- **Goal:** Compress sample approximation Q_n to reduce downstream costs
 - e.g., in heart modeling, each sample point can trigger a **1000 CPU hour** simulation
- **Stein Thinning:** Greedily minimize KSD using sample points $x_1, \dots, x_n$



- **Bonus:** Corrects for biases due to off-target sampling, tempering, approximate MCMC, or burn-in [Riabiz, Chen, Cockayne, Swietach, Niederer, Mackey, and Oates, 2022]
- **Kernel Thinning:** Compress n point summary into $\sqrt{n}$ point summary with comparable KSD [Dwivedi and Mackey, 2024]

Future Directions

Many opportunities for future development

- ① Improving scalability while maintaining convergence control
 - Subsampling of likelihood terms in $\nabla \log p$ [Gorham, Raj, and Mackey, 2020]
 - Linear time, low-rank kernels that distinguish distributions [Jitkrittum, Xu, Szabó, Fukumizu, and Gretton, 2017, Huggins and Mackey, 2018]
- ② Exploring the impact of Stein operator choice
 - An infinite number of operators $\mathcal{T}$ characterize P
 - How is discrepancy impacted? How do we select the best $\mathcal{T}$?
 - Diffusion Stein operators [Gorham, Duncan, Vollmer, and Mackey, 2019]; Best $\mathcal{T}$ for **unexplored modes**?
- ③ Addressing other inferential tasks
 - Generative models [Wang and Liu, 2016, Pu, Gan, Henao, Li, Han, and Carin, 2017, Shi, Zhou, Hwang, Titsias, and Mackey, 2022b]
 - Parameter estimation [Barp, Briol, Duncan, Girolami, and Mackey, 2019]
 - Post-selection inference [Shi, Liu, and Mackey, 2022a]
 - Non-convex optimization [Erdogdu, Mackey, and Shamir, 2018]
 - Distribution compression [Li, Dwivedi, and Mackey, 2024]
 - Control variates [Assaraf and Caffarel, 1999, Mira, Solgi, and Imparato, 2013, Oates, Girolami, and Chopin, 2016, Shi, Zhou, Hwang, Titsias, and Mackey, 2022b]

Probabilistic Inference and Learning with Stein's Method

Qiang Liu Lester Mackey Chris Oates

<https://arxiv.org/pdf/2603.07467>

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Selecting Sampler Hyperparameters

Setup [Welling and Teh, 2011]

- Consider the posterior distribution P induced by L datapoints y_l drawn i.i.d. from a Gaussian mixture likelihood

$$Y_l|X \stackrel{\text{iid}}{\sim} \frac{1}{2}\mathcal{N}(X_1, 2) + \frac{1}{2}\mathcal{N}(X_1 + X_2, 2)$$

under Gaussian priors on the parameters $X \in \mathbb{R}^2$

$$X_1 \sim \mathcal{N}(0, 10) \perp\!\!\!\perp X_2 \sim \mathcal{N}(0, 1)$$

- Draw $m = 100$ datapoints y_l with parameters $(x_1, x_2) = (0, 1)$
- Induces posterior with second mode at $(x_1, x_2) = (1, -1)$
- For range of parameters ϵ , run approximate SGLD for 1000 steps and store resulting posterior sample Q_n
- Use minimum GSD to select appropriate ϵ
 - Compare with standard MCMC parameter selection criterion, effective sample size (ESS), a measure of Markov chain autocorrelation
 - Compute median of diagnostic over 50 random sequences

Selecting Samplers

Setup

- **MNIST handwritten digits** [Ahn, Korattikara, and Welling, 2012]
 - 10000 images, 51 features, binary label indicating whether image of a 7 or a 9
- Bayesian logistic regression posterior P
 - L independent observations $(y_l, v_l) \in \{1, -1\} \times \mathbb{R}^d$ with

$$\mathbb{P}(Y_l = 1 | v_l, X) = 1 / (1 + \exp(-\langle v_l, X \rangle))$$

- Flat improper prior on the parameters $X \in \mathbb{R}^d$
- Use IMQ KSD ($\beta = -\frac{1}{2}, c = 1$) to compare SGFS-f to SGFS-d drawing 10^5 sample points and discarding first half as burn-in
- For external support, compare bivariate marginal means and 95% confidence ellipses with surrogate ground truth Hamiltonian Monte chain with 10^5 sample points [Ahn, Korattikara, and Welling, 2012]

The Importance of Tightness

Goal: Show $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$ only if Q_n converges to P

- A sequence $(Q_n)_{n \geq 1}$ is **uniformly tight** if for every $\epsilon > 0$, there is a finite number $R(\epsilon)$ such that $\sup_n Q_n(\|X\|_2 > R(\epsilon)) \leq \epsilon$
 - Intuitively, no mass in the sequence escapes to infinity

Theorem (KSD detects tight non-convergence [Gorham and Mackey, 2017])

Suppose that $P \in \mathcal{P}$ and $k(x, y) = \Phi(x - y)$ for $\Phi \in C^2$ with a non-vanishing generalized Fourier transform. If $(Q_n)_{n \geq 1}$ is uniformly tight and $\mathcal{S}(Q_n, \mathcal{T}_P, \mathcal{G}_k) \rightarrow 0$, then $(Q_n)_{n \geq 1}$ converges weakly to P .

- **Good news**, but, ideally, KSD would detect non-tight sequences automatically...